Introduction to Theoretical Computer Science

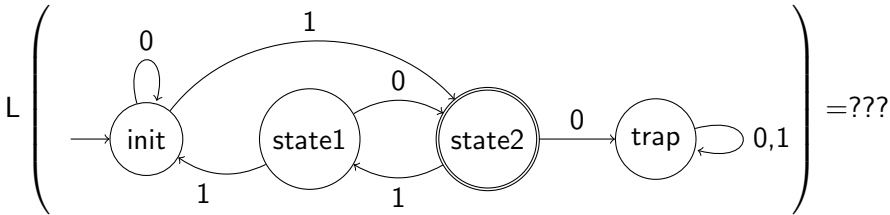
CS2490 - Session 5: Regular Languages, constructing DFAs

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Fall 2026

August 26, 2026

Venture a guess



Recall

The language accepted by a DFA $\mathcal{A} = (Q, \Sigma, \delta, q_0, F)$ is

$$L(\mathcal{A}) = \{w \in \Sigma^* \mid \delta^*(q_0, w) \in F\}$$

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Let's explore $\mathrm{L}(\mathcal{A})$ in our case!

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Spoiler: we will see a method to systematically find "good" descriptions from any automaton.

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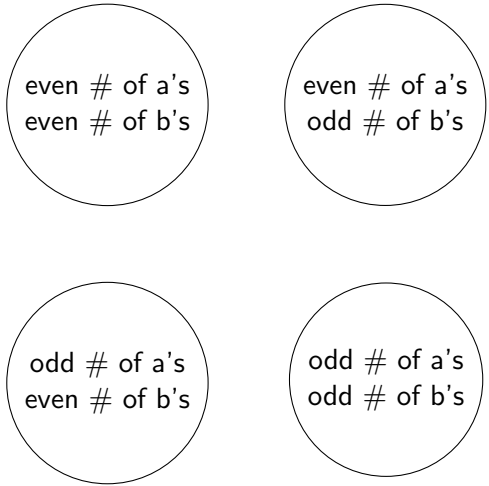
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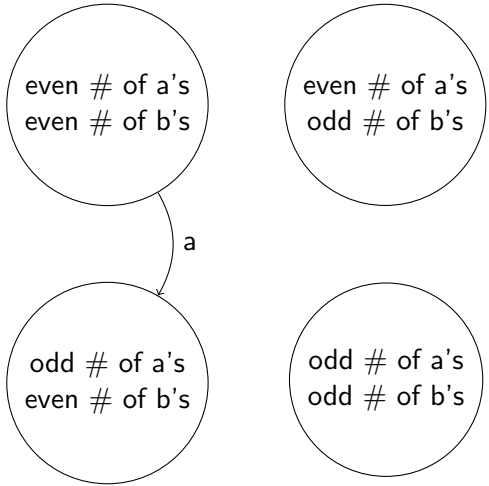
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(The alphabet here is $\{a, b\}$.)

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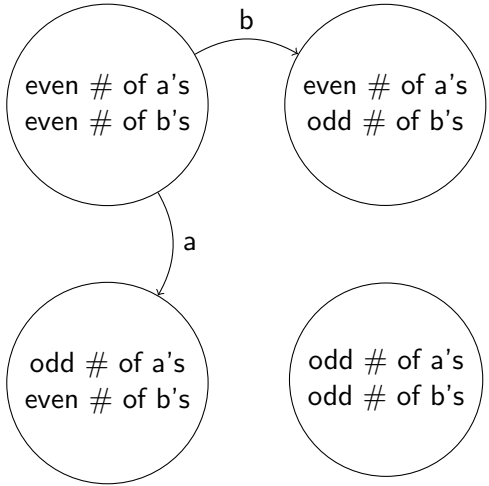
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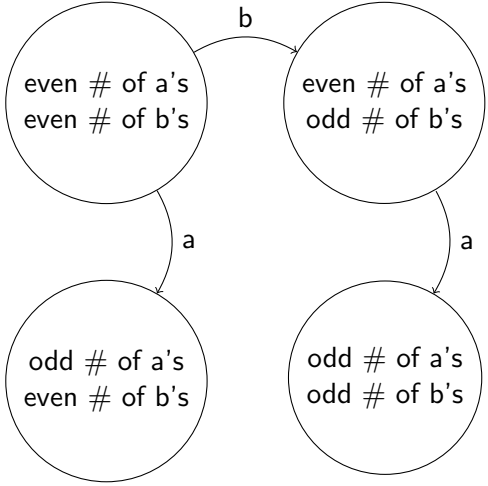
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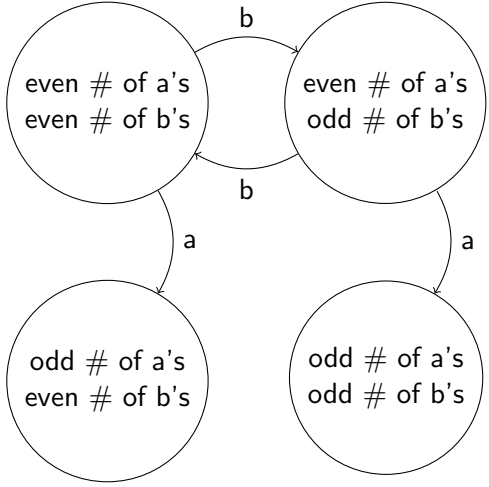
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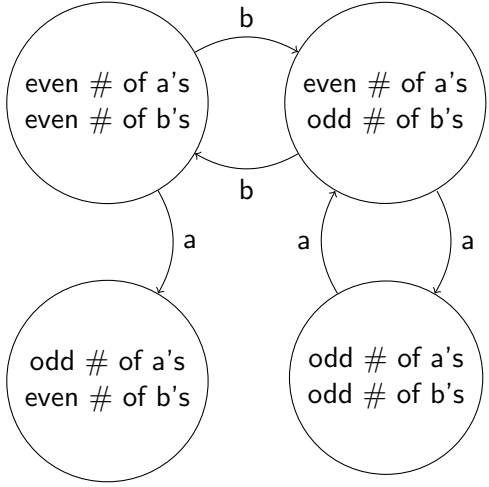
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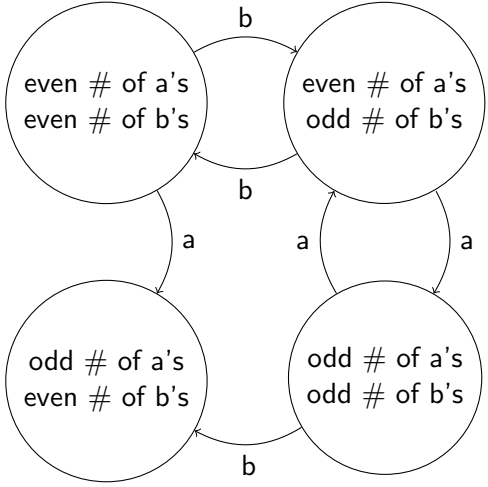
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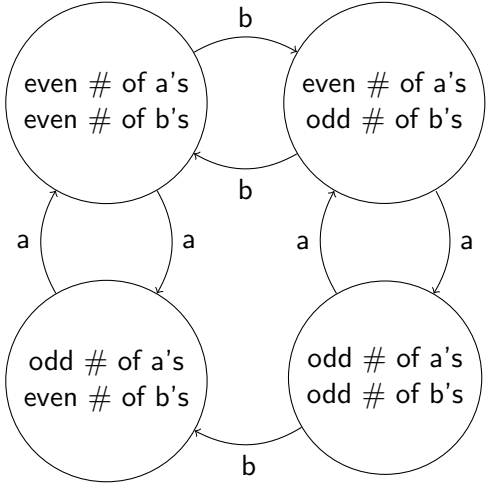
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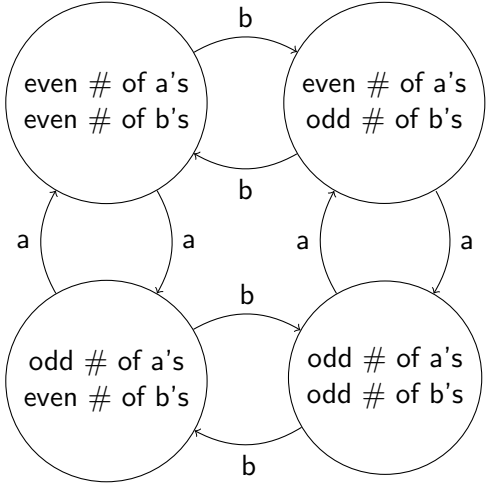
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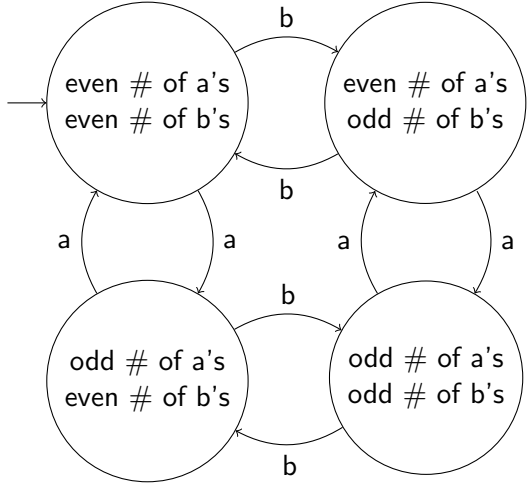
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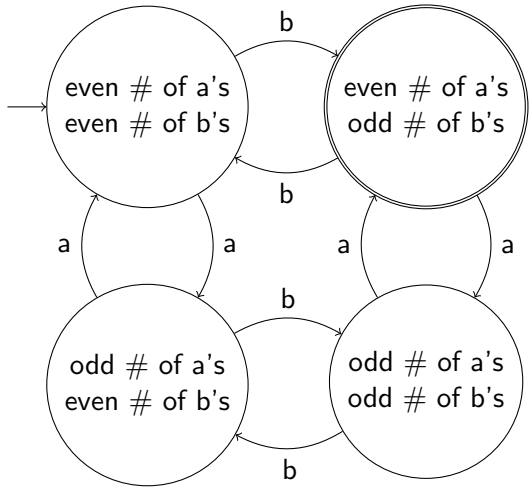
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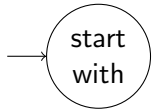
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So L can be rewritten as

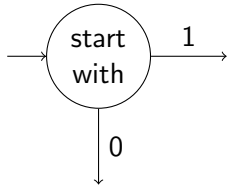
$$L = \{w \in \{0,1\}^* \mid w \text{ starts and ends with } 1\}.$$

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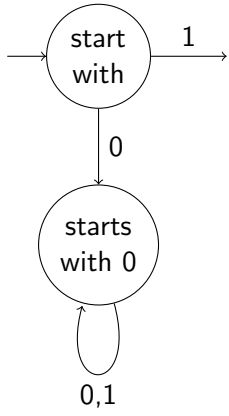
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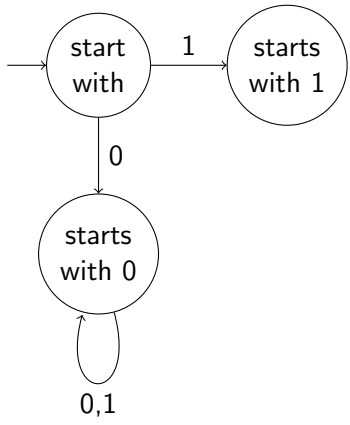
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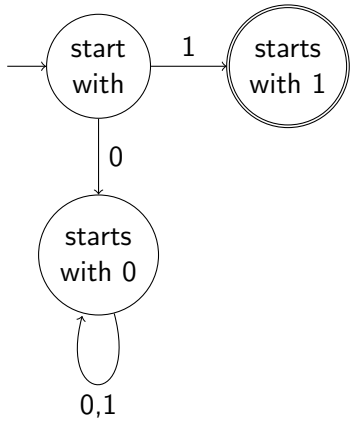
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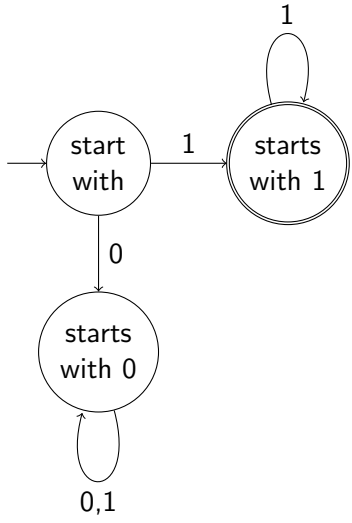
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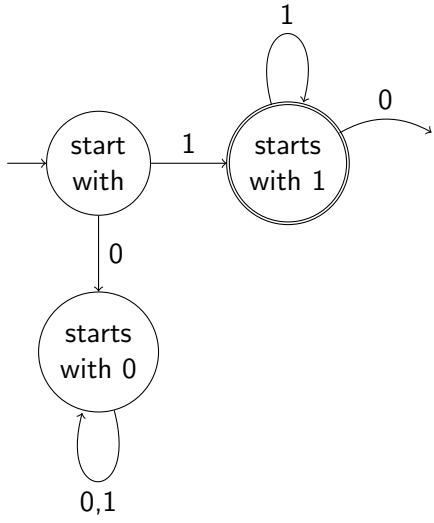
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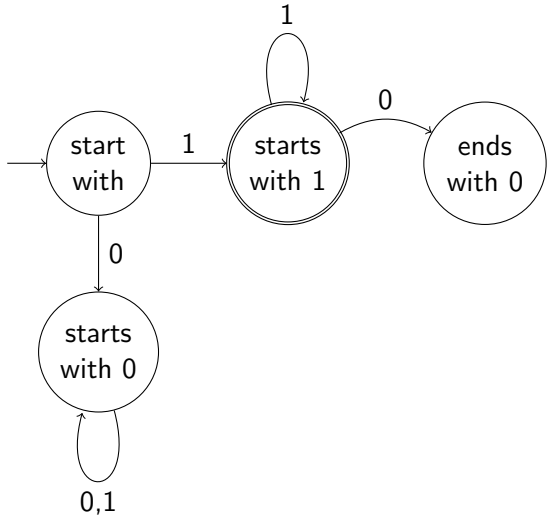
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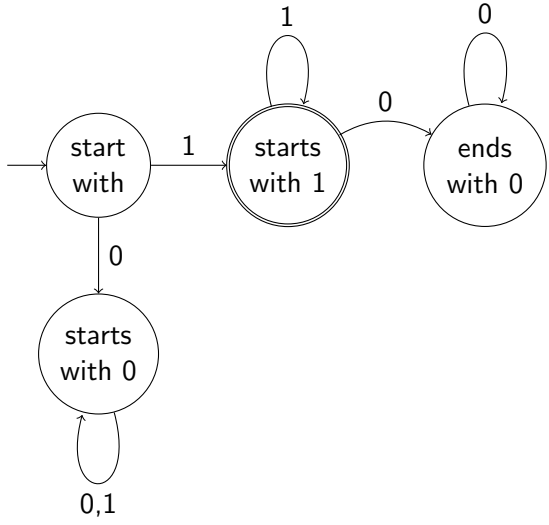
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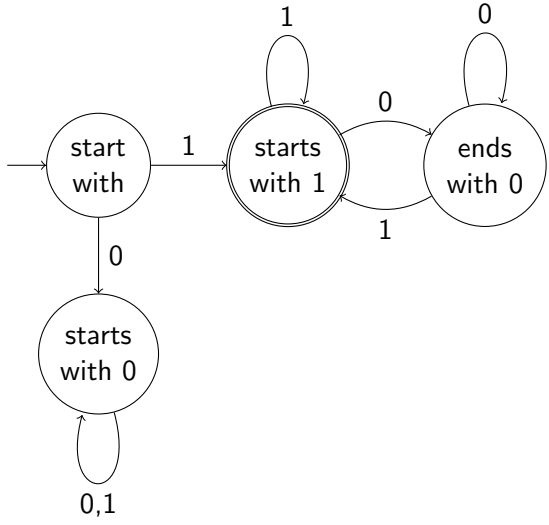
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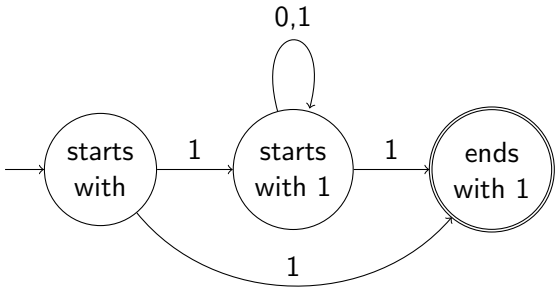
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But what about this simpler one?



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We write $\mathcal{A} = (\Sigma, Q, \delta, I, F)$.